## 5.6 Numerical Methods and Invariant Distributions

In the previous section, we introduced the temporal average of a function along a solution of a differential equation (as a means of computing the microcanonical ensemble average). Since we cannot compute an exact trajectory, we would hope that such an average could be computed in an analogous way using a numerical method. If $z_n$ represents the approximation to $z(t_n)$, then we might attempt to estimate $\bar{g}_\Gamma$ by

$$\bar{g}_\Gamma \approx \lim_{n\to\infty} n^{-1} \sum_{k=0}^{n} g(z_k).$$

However, as is well known, numerical methods may introduce growing errors so an infinite average might not make sense, even if the time average does! Still we might hope that trajectory averages could be approximated by discrete averages of moderately long length. In the several examples in this chapter, we have used numerical methods to simulate Hamiltonian dynamics in order to gain intuition about their ergodic properties, but we have not discussed the effect of the choice of the discretization method itself on this issue.

We have seen that the Liouville equation defines the propagation of the distribution associated to system of differential equations. In a similar way, a numerical method $z_{n+1} = \mathscr{F}_h(z_n)$ induces an effective distributional propagator. One way to understand this propagator is to interpret the numerical method as being approximately equivalent to the solution of a perturbed differential equation

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z) + h^r \boldsymbol{f}_{(r)}(z) \equiv \tilde{\boldsymbol{f}}_h(z).$$

The perturbation term may be viewed as the finite truncation of the perturbative expansion obtained from the backward error analysis using the methods of the previous chapter. ($r$ is the order of the numerical method.) Let us assume that the perturbed equation is a realistic model for the numerical solutions, then define the evolving density by

$$\frac{\partial \rho}{\partial t} = \mathcal{L}^\dagger_{\tilde{f}} \rho \stackrel{\text{def}}{=} \mathcal{L}^\dagger_f \rho + h^r \mathcal{L}^\dagger_{f_{(r)}} \rho.$$

Under suitable boundedness assumptions on the remainder, we could expect that the distributional error remains bounded by a quantity of order $h^r$. Unfortunately, it is not always true that $\rho$ remains bounded.

As a simple illustration, consider the harmonic oscillator with Hamiltonian $H(q,p) = p^2/2 + q^2/2$ and the invariant distributions obtained using several numerical methods. Applying Forward Euler to the harmonic oscillator, we have

$$q_{n+1} = q_n + hp_n, \qquad p_{n+1} = p_n - hq_n,$$